

Assignment 5: Teleportation Service

Consider a quantum network architecture where Applications can directly request the teleportation of an arbitrary qubit state ρ from any end node to another.

Use **QuantumSavory/Julia** to simulate the Application-level perspective of the Quantum Network (QN). Specifically:

1. The QN is represented as a set of N fully-connected end-nodes. As shown in Figure 1, a central, virtual node representing the *QN core* can be used to simplify the connectivity and manage the simulation.
2. At any time, an application can request the teleportation of an arbitrary qubit state ρ from a random end node to another.
3. Every application request is served by
 - a. delivering the qubits of a Bell pair to the two interested end nodes with a randomized delay following an exponential distribution with average time $t = 200ms$ (the *QN core* provides this functionality and abstracts all the underlying complexities such as link latencies). The generated Bell pairs are depolarized states with parametrized depolarization probability p_w .
 - b. carrying out the quantum teleportation protocol between the two end nodes, employing the qubit state ρ to teleport and the (noisy) Bell pair from the previous step.

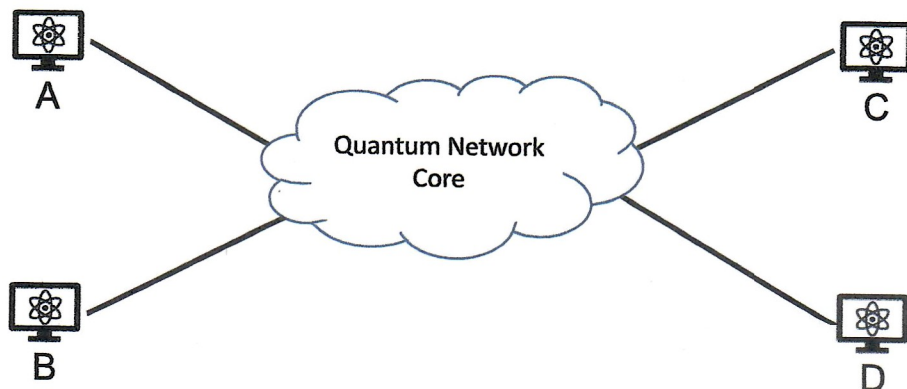


Figure 1

Implement and test the quantum network functionalities explained above. Assume that application requests are served sequentially (one request at a time). The setup shown in Figure 1 can be used as an example.

Qiskit Implementation

Consider a quantum network architecture where Applications can directly request the teleportation of an arbitrary qubit state ρ from any end node to another.

Use **Qiskit/Python** to simulate the Application-level perspective of the Quantum Network (QN). Specifically:

1. The QN is represented as a set of N fully-connected end-nodes. A central, virtual node representing the QN core abstracts all underlying network complexities and manages teleportation requests. The network topology is handled as a classical abstraction in Python, since the "fully-connected" property simply means any pair of nodes can be selected for a teleportation request.
2. At any time, an application can request the teleportation of an arbitrary qubit state ρ from a random end node to another.
3. Every application request is served by:
 - a. Generating a Bell pair and delivering it to the two interested end nodes. The generated Bell pairs are **depolarized states** with parametrized depolarization probability p_w .
 - b. Carrying out the quantum teleportation protocol between the two end nodes, employing the qubit state ρ to teleport and the (noisy) Bell pair from the previous step.
4. Application requests are served **sequentially** (one request at a time).

Deliverables

1. **Quantum Teleportation Circuit:** Implement the standard teleportation protocol with classical corrections conditioned on Bell-basis measurement outcomes.
2. **Noisy Bell Pair Generation:** Apply a depolarizing channel to the Bell pair with configurable probability p_w .
3. **Network Simulation Loop:** For a configurable number of requests, randomly select sender/receiver node pairs from the N -node network and execute the teleportation protocol.
4. **Fidelity Analysis:** Compute the teleportation fidelity $F(\rho, \rho')$ between the original state ρ and the teleported state ρ' . Analyze fidelity as a function of the depolarization probability p_w .